

Technical Report

Region of attraction with Integral Quadratic Constraints

Seminar:

System Theoretical Methods for Flight Control

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1 Introduction

1.1 General Remarks/Tips

- Using `\autoref{fig:Cmg}` to refer to labels (such as Figure 1.1) has two advantages: First, the notation will be consistent throughout the entire document. Second, the hyperlink will cover the denotation as well.

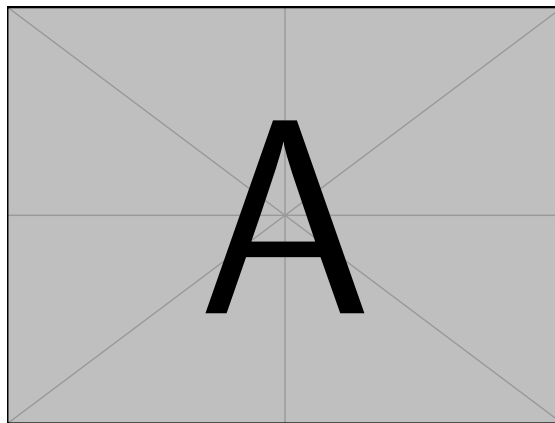


Figure 1.1: Control Moment Gyro

- The **AMSLatex** package¹ provides many useful commands for math mode, the arguably most important being the environments **equation*** for unnumbered equations, **align** and **align*** for arrays of equations (the starred version has no numbers), and **pmatrix** or **bmatrix** for matrices with parentheses or brackets, respectively:

$$\begin{aligned}\dot{x} &= A(\rho) x \\ &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}(\rho) \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\end{aligned}\tag{1.1}$$

- Using the **acronym** package², acronyms defined in the **acronym** environment can be used with `\ac` and other commands (see Table 1.1)

¹<https://ctan.org/pkg/amslatex>

²<https://ctan.org/pkg/acronym>

Table 1.1: Acronym Commands

What?	Command	Output
First call	<code>\ac{UAV}</code>	UAV! (UAV!)
All following calls	<code>\ac{UAV}</code>	UAV!
With indef. article	<code>\Iac{UAV}</code> , <code>\iac{UAV}</code>	A UAV! , a UAV!
Plural	<code>\acp{UAV}</code>	UAV! s
Use long form again	<code>\acl{UAV}</code>	UAV!

- The `biblatex` package³ provides several useful citation commands in addition to `\cite`. For example, `\textcite{Notter2020MultipleThermalUpdraft}` yields Notter et al. [1]
- A nice feature of the `siunitx` package⁴ is the definition of table columns for numbers such that they are aligned at the decimal marker (see Table 1.2).

Table 1.2: Numbers and Units in Tables

	Parameter	Value	
Position	p	120.4	m
Time	t	0.08	s

1.2 Typesetting Formulas

This is a brief summary of typographical conventions for technical/scientific writing based on DIN 1338. As a general rule of thumb, **variables and physical quantities are set in *italic font*; everything else is set in upright font**. See Table 1.3 and the following list for rules and examples.

Table 1.3: Typesetting Formulas

What?	How?	Examples
Numbers denoted by digits	upright	1, 2, 3
Numbers denoted by symbols	<i>italic</i>	x, y, z, i, j, k
Mathematical constants	upright	e, π , j
Physical constants (e.g. speed of light)	<i>italic</i>	c
Units (including their prefixes)	upright	m, km, μ rad
Indices follow the same convention	upright/ <i>italic</i>	$x_{\text{true}} = \sum_{i=1}^n y_i$
Functions with arbitrary symbol	<i>italic</i>	$f(x), g(u)$
Conventional functions	upright	sin, cos, ln, exp

³<https://ctan.org/pkg/biblatex>

⁴<https://ctan.org/pkg/siunitx>

- Conventional functions and operators are separated from their arguments by a small space, e.g. $\sin \phi$, $\ln x$. Exceptions are the differential and difference operators, i.e. ∂x , $d\phi$, and Δt . $\text{\LaTeX}$ usually gets this right, so there should be no need to include $\backslash,$ between function and argument.
- Products are denoted by setting the factors next to each other ($x = yz$); potential ambiguities can be resolved by setting a small space with $\backslash,$ ($x = y\,z$).
- Note the difference in German and English typesetting of numbers:

German: $1,234 \cdot 10^{-5}$

English: 1.234×10^{-5}

The `\num` command of the **siunitx** package⁵ takes care of this: `\num{0,12e3}` and `\num{.12e3}` yield 0.12×10^3 and 0.12×10^3 .

- Numbers and their units (except the degree symbol) are separated by a small space. The easiest way is to use the **siunitx** package: For instance, `\qty{12345,67}{km}` and `\qty{90}{\degree}` yield 12 345.67 km and 90° , respectively.

⁵<https://ctan.org/pkg/siunitx>

Bibliography

- [1] Stefan Notter, Pascal Groß, Philipp Schrapel, and Walter Fichter. “Multiple Thermal Updraft Estimation and Observability Analysis”. In: *Journal of Guidance, Control, and Dynamics* 43.3 (Mar. 2020), pp. 490–503. DOI: 10.2514/1.g004205.
- [2] Ramin Geshnizjani, Andrei Kornienko, Tobias Ziegler, Johannes Löhr, and Walter Fichter. “Torque Optimal Steering of Control Moment Gyroscopes for Agile Spacecraft”. In: *Journal of Guidance, Control, and Dynamics* 44.3 (Dec. 2021), pp. 629–640. DOI: 10.2514/1.g005118.
- [3] Lorenz Schmitt and Walter Fichter. “Cramér-Rao Lower Bound for State-Constrained Nonlinear Filtering”. In: *IEEE Signal Processing Letters* 24.12 (Dec. 2017), pp. 1882–1885. DOI: 10.1109/lsp.2017.2764540.